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How pH and sulfation shape human heparanase structure, stability, and substrate recognition

Angela Pennacchio [a,1], Yesid Aristizabal [b,1], José Oñate-Garzón [b], Sabato D'Auria [a,*], Alessio Ausili [c,**]

[a] Istituto di Scienze dell'Alimentazione, CNR, Avellino, 83100, Italy
[b] Grupo de Investigación en Química y Biotecnología (QUIBIO), Facultad de Ciencias Básicas, Universidad Santiago de Cali, Cali, Colombia
[c] Chemical Engineering Department, Faculty of Chemistry, Regional Campus of International Excellence "Campus Mare Nostrum", University of Murcia, Murcia, 30071, Spain

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ABSTRACT

Human heparanase (HPSE) is an endo-β-D-glucuronidase involved in the degradation of heparan sulfate and in several physiopathological processes, including inflammation, metastasis, and extracellular matrix remodeling. Its activity is strongly influenced by pH and by the sulfation degree of the substrate, yet the structural bases underlying these effects remain only partially understood. In this work, we combine FTIR spectroscopy, temperature-dependent intrinsic fluorescence, and molecular dynamics (MD) simulation experiments to investigate the secondary structure, conformational stability, and structural determinants of substrate recognition in HPSE under different physicochemical conditions.

FTIR analyses reveal pH-dependent variations in secondary structure, with acidic environments promoting a more compact and α-helix-rich conformation, consistent with the lysosomal conditions in which HPSE is physiologically active. Fluorescence experiments show a gradual unfolding process at all pH values, with maximal protein stability at pH 5.0.

MD simulations of HPSE complexed with heparan-sulfate tetrasaccharides of different sulfation degrees indicate that ionic and hydrogen-bond interactions strongly contribute to binding affinity with highly sulfated ligands forming more stable and persistent interactions.

Overall, the integration of spectroscopic analyses and computational simulations suggests that acidic pH and high substrate sulfation favor structural stability and ligand binding in HPSE, providing insights into the physicochemical factors regulating its activity.

1. Introduction

The mammalian enzyme heparanase (HPSE) is an endo-β-D-glucuronidase belonging to the GH79 family, responsible for cleaving β(1 → 4) linkages in heparan sulfate glycosaminoglycan (HS-GAG) chains of heparan sulfate proteoglycans (HSPG) [1]. The enzyme is synthesized as an inactive 65 kDa precursor (pro-HPSE) that, upon proteolytic processing, yields the catalytically active heterodimeric form composed of non-covalently associated 8 kDa and 50 kDa subunits [2–6]. Recombinant production of the two subunits enabled reconstitution of active HPSE and investigation of its substrate specificity through computational approaches [7].

HPSE plays a crucial role in remodeling the extracellular matrix (ECM) and basement membrane, promoting processes such as angiogenesis, inflammation, and tissue repair through the controlled degradation of HS-GAG chains [3,8,9]. Cleavage of heparan sulfate chains also facilitates the release of growth factors, chemokines, and cytokines bound to HSPGs, thereby influencing cell proliferation, migration, and differentiation [10]. HPSE overexpression has been correlated with the invasive and metastatic behavior of various tumor cell lines [11–14] making the enzyme a promising target for the anti-metastatic and anti-angiogenic therapies [15–18].

* Corresponding author.
** Corresponding author.
E-mail addresses: sabato.dauria@cnr.it (S. D'Auria), aausili@um.es (A. Ausili).
[1] These authors contributed equally to the work.

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The catalytic activity of heparanase is strongly pH-dependent, showing maximal activity under acidic conditions and decreasing with increasing alkalinity [19–21]. This behavior reflects the ionization state of catalytic residues, particularly Glu343 and Glu225, which act as the nucleophile and acid/base, respectively, in the hydrolysis mechanism [22]. The optimal activity of HPSE in acidic environments is consistent with its lysosomal localization, where pro-HPSE maturation and glycosaminoglycan degradation occur. In pathological contexts such as solid tumors, acidic extracellular pH can mimic lysosomal conditions, allowing heparanase to remain active outside the cell [23]. Protonation of catalytic and basic residues may favor catalytically competent conformations, whereas neutral pH may weaken these interactions and reduce enzymatic efficiency [24,25].

Structurally, HPSE consists of a catalytic $(\beta/\alpha)_8$ domain and a β-sandwich domain involved in stability and substrate recognition [22]. The 8 kDa subunit contributes to both domains, while the larger subunit completes the catalytic structure [22,26,27]. A cluster of basic residues stabilizes substrate sulfate groups, conferring specificity for highly sulfated regions [28,29].

HPSE selectively recognizes disaccharide units containing 3-O- or 6-O-sulfate groups and shows preference for 2-O-sulfated glucuronic acid over iduronic acid residues [30–32]. Heparan sulfate chains consist of repeating disaccharide units of uronic acid (GlcUA or IdoUA) and glucosamine (GlcN) whose post-synthetic modifications such as N-sulfation, O-sulfation, and epimerization generate structurally distinct NS (sulfate-rich) and NA (non-sulfated) domains [33,34]. This variability modulates interactions with numerous biological ligands, including growth factors, cytokines, and ECM proteins [30,35].

HPSE affinity for HS substrates increases with sulfation degree, as sulfate groups stabilize interactions with the basic residues of the catalytic site through ionic and hydrogen bonds [35–38]. Molecular dynamics simulations have shown that highly sulfated ligands form more stable complexes than less sulfated tetrasaccharides [39,40].

The combined effect of pH and sulfation therefore represents key regulatory factors for HPSE activity. While pH modulates the protonation state of catalytic and charged residues, sulfation density influences the strength of enzyme–substrate interactions.

In this study, we provide an integrated analysis of the factors modulating HPSE structure, stability, and substrate recognition by combining experimental and computational approaches. FTIR spectroscopy and temperature-dependent fluorescence were performed on the apo enzyme to evaluate the effect of pH on secondary structure and thermal stability. In parallel, molecular dynamics simulation experiments with three heparan sulfate tetrasaccharides of increasing sulfation degree were used to investigate how pH and ligand sulfation modulate HPSE–substrate interactions at the atomic level. Although substrate was not included in the spectroscopic experiments, the combination of apoprotein measurements and ligand-bound simulations provids a coherent framework for understanding the structural and chemical determinants regulating HPSE activity.

2. Materials and Methods

2.1. Materials

Deuterium oxide (99.9% ^2H_2O), ^2HCl, NaO^2H, were purchased from Sigma-Aldrich. Tris-base and sodium acetate anhydrous were obtained from Fisher Scientific. All other reagents and solvents were commercial samples of the highest purity.

HPSE was expressed and purified as previously described [7].

2.2. Sample preparation for FTIR measurements

An amount of 1.2 mg of protein dissolved in storage buffer was used to prepare each sample to be measured by means of Fourier transform infrared spectroscopy (FTIR). The protein was analyzed in three different buffers prepared in ^2H_2O at different values of p^2H: 50 mM sodium acetate, p^2H 3.5; 50 mM sodium acetate, p^2H 5.0; 50 mM Tris, p^2H 6.5. In order to completely replace the buffer used for the protein purification with the deuterated ones, the protein was firstly concentrated into an approximate volume of 50 μl by centrifugation in a Vivaspin 500 Centrifugal Concentrator with 10,000 MWCO PES membrane (Sartorius) at 10,000 g and at 4 °C, then 200 μl of deuterated buffers was added and the sample was concentrated again. This washing process was repeated 6 times and the in last one, which took 15 h, the volume was reduced to about 40 μl for the FTIR measurements.

2.3. Collection and analysis of FTIR spectra

The concentrated protein sample was placed in a standard rectangular cell equipped with CaF_2 windows and 25 μm Teflon spacers (PerkinElmer). FTIR spectra were recorded by means of a PerkinElmer Frontier MIR/FIR Spectrometer using a deuterated triglycine sulfate (DTGS) detector and a normal Beer-Norton apodization function. A total of 64 scans were averaged for each spectrum with a nominal resolution of 2 cm^{-1} and before every experiment, background single-beam spectra were collected. At least 4 h before and during data acquisition, the spectrometer was continuously purged with a stream of N_2. Samples and buffers were scanned at room temperature (RT).

All the spectra were collected and processed using the Spectrum software from PerkinElmer. ^2H_2O buffer spectra were subtracted from the respective sample spectra. Second derivative spectra were calculated over a 19 data-point range (19 cm^{-1}) and the parameters of the deconvoluted spectra were set with a γ value of 2.5 and a smoothing length of 55.

The quantitative analysis of the protein secondary structure was performed by curve-fitting of the deconvoluted amide I' band [41] using the peak fitting module of the Origin 8.5 software (OriginLab Corporation, Northampton, MA 01060 USA). The band shape was set to a Gaussian curve, and the fitting was obtained by iteration in two steps, the first iteration was performed fixing the peak positions as obtained by second derivative and deconvolution, while in the second step the bands were free [42].

2.4. Steady-state fluorescence measurements

Steady-state fluorescence emission experiments were performed on ISS-K2 Frequency-Domain Fluorometer (ISS, Urbana-Champaign) equipped with a cell temperature-controlled sample holder. To selectively excite Trp residues, the excitation wavelength was set at 295 nm. Excitation and emission slit widths were fixed at 0.5 nm and 1.0 nm respectively. Emission spectra recorded from 310 nm to 410 nm, at 1.0 nm intervals, with a scan speed of 100 nm/min and lamp power at 17.5 A. Samples were placed in the thermostatic holder and the temperature of samples was measured directly in a 1.0 cm light path quartz cuvette with four optical faces. Fluorescence experiments were performed on recombinant HPSE sample with optical density lower than 0.1 OD at 295 nm to minimize inner filter effects. HPSE thermal stability was analyzed in three different buffers at different values of pH: 50 mM sodium acetate, pH 3.5, 50 mM sodium acetate, pH 5.0 and, 50 mM Tris, pH 6.5 in a final volume of 500 μL. The buffer was used as blank, and its emission contribution was subtracted from the spectra.

To analyze the thermal proteins stability, studies on temperature-induced denaturation were conducted of the HPSE. Fluorescence intensity variation on temperature was recorded in the range 25-95 °C. Measurements were performed in triplicate after 5 min of incubation at defined temperature. Three independent experiments were carried out. The values were reported as the means ± standard deviation. Thermal denaturation curves were obtained by plotting the fluorescence intensity at the emission maximum against temperature. The melting temperature (T_m) was estimated by fitting the denaturation curves to a Boltzmann sigmoidal model, and T_m was defined as the inflection point of the fitted

curve. Curve fitting and data analysis were performed using OriginPro 2021b software.

2.5. Molecular dynamics simulations

Molecular dynamics (MD) simulations were carried out to evaluate the structural stability of heparanase and its interaction with oligosaccharide ligands under different pH conditions. Three heparan sulfate (HS) tetrasaccharides differing in their sulfation pattern were considered: (i) HS-tetra(0S), non-sulfated [GlcNAc-α1,4-GlcUA-β1,4-GlcNAc-α1,4-GlcUA]; (ii) HS-tetra(NS), N-sulfated at both glucosamine units [GlcNS-α1,4-GlcUA-β1,4-GlcNS-α1,4-GlcUA]; and (iii) HS-tetra(NS,2S), N-sulfated at glucosamines and 2-O-sulfated at uronic acids [GlcNS-α1,4-GlcUA-2S-β1,4-GlcNS-α1,4-GlcUA-2S]. The set of tetrasaccharide ligands used in the simulations was selected to primarily explore the effect of increasing sulfation density on the interaction with HPSE. Although heparanase substrate recognition also depends on specific saccharide sequence motifs, the present study focuses on the contribution of electrostatic interactions associated with different sulfation levels while maintaining a comparable backbone structure. For each ligand, independent simulations were performed at three protonation states corresponding to pH 3.5, 5.0, and 6.5.

Initial protein–ligand complexes were built based on the crystallographic structure of HPSE (PDB: 5E8M) and docked oligosaccharides. All molecular dynamics simulations were performed using the CHARMM36 m force field, which was applied consistently to protein, carbohydrate ligands, water, and ions. Protonation states of titratable residues were set according to the target pH, and each system was solvated with explicit TIP3P water molecules and neutralized with 0.15 M KCl [43]. Molecular dynamic simulations were performed with GROMACS software version 2021.1 [44]. After energy minimization, the systems were gradually heated to a temperature of 310 K with an integration step of 1 fs (femtoseconds) for 75 ps (picoseconds). For the equilibration phase, 300 ps were performed using a 2 fs step. Once the system was equilibrated at the desired temperature and pressure, molecular dynamics simulations were carried out for the production stage at a maximum time of 100 ns. Long-range electrostatic interactions were treated using the Particle Mesh Ewald (PME) method [45], and bonds involving hydrogen atoms were constrained with the SHAKE algorithm, allowing the use of a 2 fs integration step [46].

Analyses were performed using PyMol [47], BIOVIA Discovery Studio 2024 Client [48] GROMACS and Python. For energy calculations, gmx_MMPBSA [49] was used, a tool based on AMBER's MMPBSA.py that aims to perform final state free energy calculations with the simulation path files obtained in GROMACS. Structural stability was monitored by calculating the root mean square deviation (RMSD) of Cα atoms, while overall equilibration of the system was evaluated from the total potential energy expressed in kcal/mol. The evolution of the secondary structure was determined using the STRIDE algorithm [50]. Protein–ligand interactions were examined by calculating hydrogen bond occupancy and contact frequencies between ligand groups and protein residues, classifying the interactions as strong (+2), moderate (+1), weak (−1), or transient (−2) according to their persistence throughout the trajectory.

3. Results

3.1. Heparanase secondary structure

Fig. 1 shows the absorbance (A), the deconvoluted (B) and the second derivative (C) spectrum of HPSE at p^{2}H 5.0 and RT. Resolution enhanced spectra (B and C), obtained from the absorbance spectrum, were used to identify and assign the bands associated to secondary structure elements or amino acids side chains. In particular, by the examination of the amide I′ band (1700-1600 cm^{-1}) was possible to get information on the secondary structure [41,51]. In the case of HPSE,

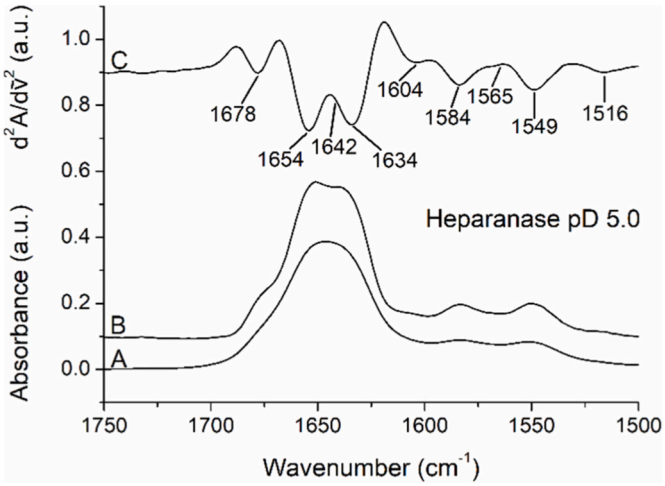


Fig. 1. Original absorbance (A), deconvoluted (B) and second derivative (C) FTIR spectra of HPSE between 1750 and 1500 cm^{-1} at RT. The sample was measured in ^2H_2O. The buffer was 50 mM sodium acetate, p^{2}H 5.0.

four main bands can be detected and associated to specific secondary structure elements (Fig. 1C). The main peak of the amide I' band located at 1654 cm^{-1} is due to α-helix structures and indicates that at p^{2}H 5.0 this component is predominant, whereas the presence of β-sheet structure is associated to the characteristic band arising at 1634 cm^{-1} [52]. The band at 1642 cm^{-1} is assigned to unordered structures while the band at 1678 cm^{-1} can be due to both β-turns and loops. The other bands arising below 1620 cm^{-1} are assigned to the residual amide II band (peak at 1549 cm^{-1}) and to amino acid residues. In particular, the 1604, 1584, 1565 and 1516 cm^{-1} bands are due to arginine, aspartic acid, glutamic acid and tyrosine, respectively [53].

3.2. Effect of pH on protein structure

In order to evaluate the effect of the pH on the structure of HPSE, we compared the second and deconvolved derivative spectra of the protein at three different values of p^{2}H, 3.5, 5.0 and 6.5 (Fig. 2). From the comparison, it emerges the existence of interesting changes of some structural elements. In particular, at p^{2}H 6.5, the protein undergoes an evident loss of organization in the secondary structure, as can be deduced from the decrease of the bands associated to α-helix and β-sheets (Fig. 2A), and the consequent formation of unordered structures, as it is especially evident by the shift of the amide I' band maximum (Fig. 2B) towards the spectral region characteristic of random coils absorbance (from 1652 cm^{-1} to 1642 cm^{-1}). Moreover, the decrease of the residual amide II band indicates that, at p^{2}H 6.5, a further hydrogen/deuterium exchange occurs if compared to the lower p^{2}H spectra. In fact, the relative intensity of the residual Amide II band drops to around 9.5% of the Amide I band at p^{2}H 6.5, whereas it remains markedly higher at acidic conditions, with values close to 22% and 24% at p^{2}H 5.0 and 3.5, respectively. This decrease reflects an increase in protein flexibility and/or a partial loss of native structure, also evidenced by the reduction of α- and β-structures and the concomitant rise of unordered elements. On the other hand, although there are no major differences between the spectra at p^{2}H 3.5 and 5.0, the slightly higher proportion observed at p^{2}H 3.5 suggests a more rigid and less flexible conformation of the protein under more acidic conditions (Fig. 2B).

Going into more detail, the content of the secondary structure of HPSE was estimated by using a curve fitting procedure of the deconvoluted amide I' band. In this way, it was possible to perform a quantitative comparison of the protein structure at the three values of p^{2}H studied (Fig. 3). At p^{2}H 5.0, the secondary structure composition was found to be 28% β-sheet, 37% α-helix, 25% unordered structures and

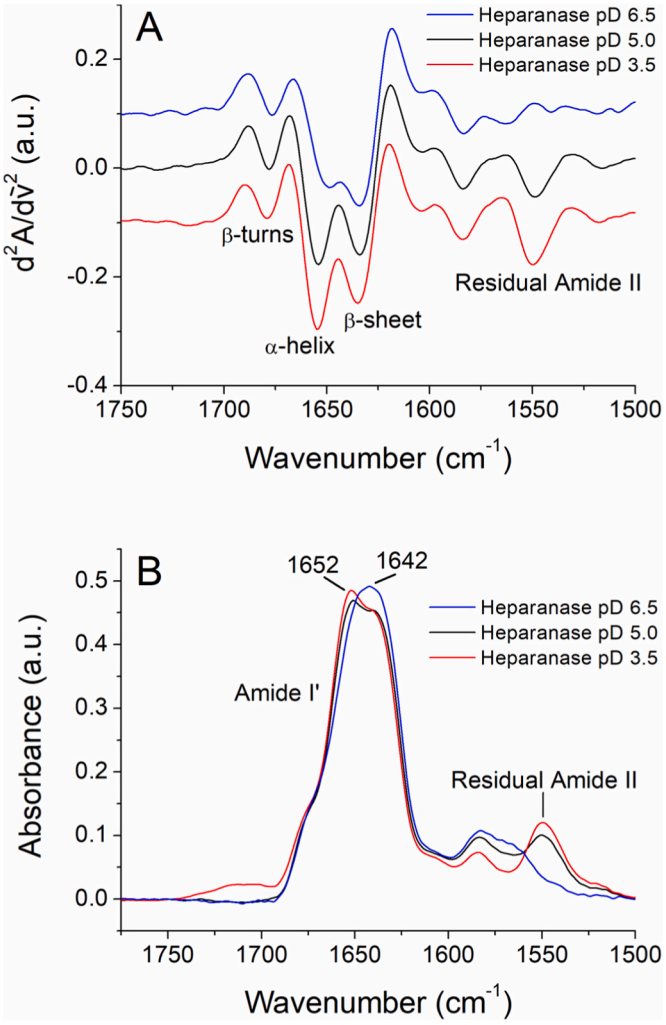


Fig. 2. Stacked second derivative (A) and deconvoluted spectra (B) at different values of p^{2}H. The spectra are reported in the 1750–1500 cm^{-1} spectral range for second derivative spectra and in the 1775–1500 cm^{-1} spectral range for deconvoluted spectra.

10% β-turns/bends. These values are in a good agreement with the X-ray data of the protein [22], with some small differences especially in the estimation of α-helix and β-turns/bends structures, probably due to the different techniques and conditions used in the measurements.

As already observed in the second derivative and deconvoluted spectra, at more neutral p^{2}H, the protein structure undergoes a partial unfolding and the curve-fitting results confirm it. In fact, by increasing the p^{2}H from 5.0 to 6.5, there is a loss of α-helix and β-sheet structures from 37% to 29% and from 28% to 23% respectively, and an increase of unordered structures from 25% to 41%. On the contrary, at a lower value of p^{2}H, there are no noticeable protein structural changes, although it is possible to observe an increase of α-helix content from 37% to 42%. This small variation, however, may still be relevant, as it may reflect subtle changes in protein structural flexibility, as shown in Fig. 2B. The results of the curve-fitting at different p^{2}H are reported in Table I.

Overall, this analysis provides insights into the structural behavior of HPSE under different pH conditions. FTIR data show that near-neutral pH destabilizes the secondary structure, leading to partial unfolding, whereas more acidic conditions promote a slightly more compact and rigid arrangement. These results describe the structural response of the apo protein to pH variations, although their direct impact on enzymatic activity cannot be inferred from the present data.

3.3. HPSE thermal stability

Thermal stability is a key parameter for understanding protein functionality, structural integrity under different conditions, and overall biological role [54]. Thermal degradation generally leads to loss of activity by disrupting the active site, altering molecular interactions, and promoting aggregation. These insights are also relevant for optimizing storage conditions and guiding protein engineering strategies aimed at improving stability [55].

The thermal effect on HPSE structure in three different buffers was evaluated by monitoring intrinsic fluorescence between 25 °C and 95 °C, using excitation at 295 nm to selectively probe tryptophan residues. In all protein samples, the emission spectra (310–410 nm) show a gradual decrease in fluorescence intensity with increasing temperature, consistent with the protein unfolding process (Fig. 4). As shown in Fig. 4, the denaturation process proceeds gradually at all pH values. While some temperature ranges show more pronounced changes, the overall decrease in fluorescence emission occurs continuously, indicating a generally gradual unfolding mechanism. However, the onset of more evident spectral changes differs among the buffers: at pH 3.5 and pH 5.0, noticeable variations appear around 62 °C, whereas at pH 6.5 they occur earlier, between 53 and 57 °C.

To obtain a clearer and comparative representation of the unfolding process, the temperature-dependent fluorescence intensities values

Table 1
Secondary structure composition of HPSE at different values of p^{2}H as calculated by curve fitting of the deconvoluted amide I′ band at RT.

Band assignment	Center (cm^{-1})	Area (%)		
		p^{2}H 3.5	p^{2}H 5.0	p^{2}H 6.5
β-sheet	~1633	27	28	23
Unordered	~1643	23	25	41
α-helix	~1656	42	37	29
β-turns/bends	~1676	8	10	7

Deconvoluted spectra were used to calculate the band position and the fractional area component bands. χ2 values for HPSE fittings are 7.8×10^{-6}, 5.9×10^{-6} and 8.0×10^{-6} at p^{2}H 3.5, 5.0 and 6.5, respectively. The values were rounded off to the nearest integer.

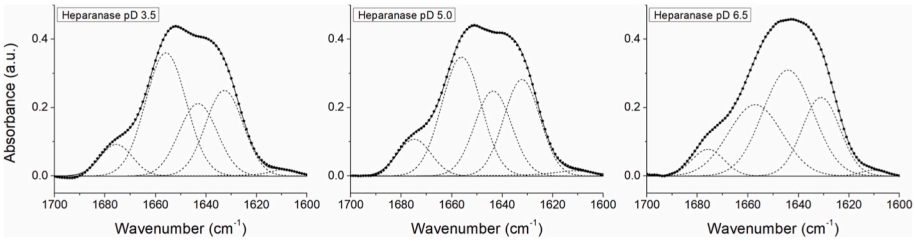


Fig. 3. Amide I′ band decomposition of HPSE at p^{2}H 3.5, 5.0 and 6.5. The spectra were Fourier self-deconvoluted before curve-fitting. Dashed lines are the individual Gaussian bands fitted to the spectrum. The dotted lines represent the original deconvoluted amide I′ bands and the continuous lines are the sum of the individual components. The numerical values are reported in Table 1.

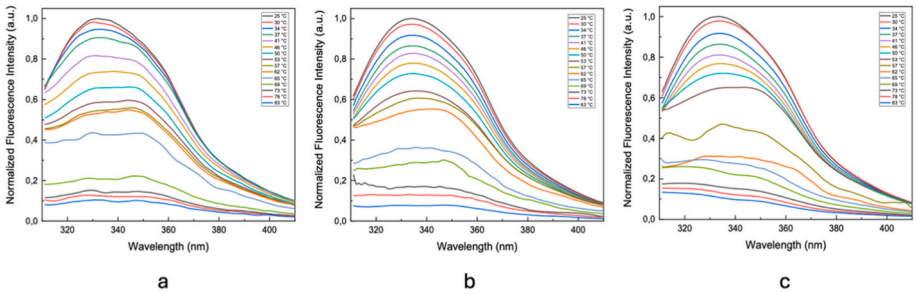


Fig. 4. Variation fluorescence emission spectra of HPSE in the temperature range from 25 °C to 95 °C at different values of pH: 50 mM sodium acetate, pH 3.5 (a); 50 mM sodium acetate, pH 5.0 (b); 50 mM Tris, pH 6.5 (c).

collected at 333 nm were plotted and fitted using a sigmoidal denaturation model, as described in material and methods section. Fig. 5 shows the denaturation curves with their corresponding Boltzmann fits, which allowed the determination of the melting temperatures (T_m), namely 59 °C, 63 °C, and 55 °C at pH 3.5, 5.0, and 6.5, respectively. These values, although with some minor apparent differences, are in agreement with the spectral observations and reflect the distinct stability regimes under the tested conditions.

Comparison of the T_m values indicates that HPSE displays the highest thermal stability at pH 5.0, with slightly lower but still elevated stability at pH 3.5. In contrast, the protein is less stable at pH 6.5, where unfolding begins at lower temperatures and the T_m is reduced. This suggests that acidic conditions favour a more rigid and compact HPSE conformation compared to near-neutral pH.

Overall, the fluorescence experiments indicate that acidic environments stabilize HPSE by promoting interactions that reduce conformational flexibility and delay the exposure of hydrophobic regions, resulting in higher T_m values. In contrast, at pH 6.5 the protein is more susceptible to thermally induced destabilization. These results, highlight that HPSE maintains its structural integrity more effectively under acidic conditions, which may be relevant for a better understanding its behaviour in physiological and pathological environments.

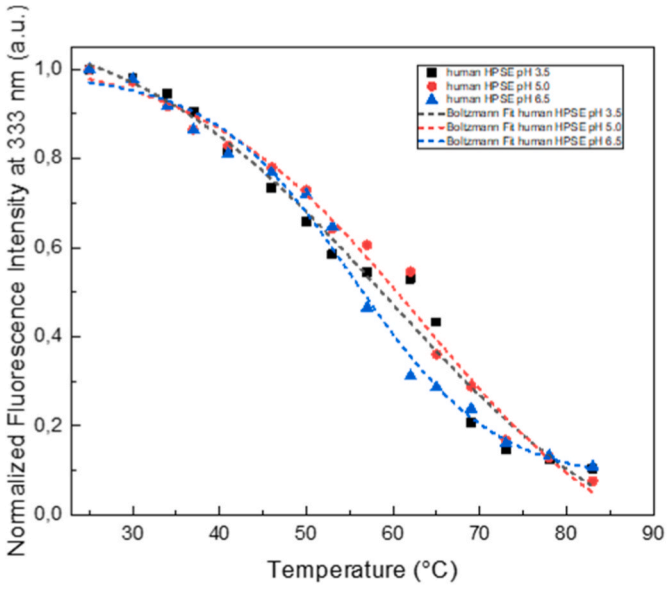


Fig. 5. Temperature-dependent fluorescence emission intensity of HPSE at 333 nm between 25 °C and 95 °C in three different buffers. Experimental data points are shown together with their corresponding sigmoidal Boltzmann fits, displayed in matching colors: 50 mM sodium acetate, pH 3.5 (black symbols and curve); 50 mM sodium acetate, pH 5.0 (red symbols and curve); 50 mM Tris, pH 6.5 (blue symbols and curve). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.4. Energy and RMSD analysis of HPSE–ligand complexes

To gain molecular-level insight into the conformational behavior of HPSE and its interaction with its ligands (heparan sulfate oligosaccharides), a series of molecular dynamics (MD) simulation experiments were performed under different protonation conditions and ligand sulfation patterns. Three heparan sulfate tetrasaccharides with increasing degrees of sulfation – HS-tetra(0S), HS-tetra(NS), and HS-tetra(NS,2S) – were examined at pH 3.5, 5.0, and 6.5.

The potential energy profiles of HPSE–substrate complexes revealed clear differences depending on pH and ligand sulfation (Fig. 6, top panels). At pH 3.5, the non-sulfated HS-tetra(0S) increased from ~ −35 to ~ −5 kcal/mol within 20 ns and stabilized near −10 kcal/mol, whereas HS-tetra(NS) decreased from −50 to ~ −75 kcal/mol in the first 5 ns and remained at this lower basin throughout the trajectory. For HS-tetra(NS,2S), energy rose from −40 to ~ −20 kcal/mol within 10 ns and then remained stable. At pH 5.0, sulfurous systems are stabilized rapidly between −15 and −10 kcal/mol, meanwhile HS-tetra(0S) showed broad oscillations (−20 to +5 kcal/mol). At pH 6.5, larger fluctuations were observed: HS-tetra(0S) shifted from −35 to ~ 0/–10 kcal/mol after 65 ns, HS-tetra(NS) fluctuated between −25 and −5 kcal/mol, and HS-tetra (NS,2S) rose from −65 to −20 kcal/mol before decreasing again to ~ −30 kcal/mol. These profiles indicate that acidic conditions favor more stable states, particularly for N-sulfated ligands, while near-neutral pH promotes higher energy and conformational instability, partially modulated by the degree of sulfation.

The RMSD of the protein backbone (Cα atoms) revealed overall stable trajectories with gradual deviations depending on pH and ligand type (Fig. 7, middle panels). In the presence of HS-tetra(0S), RMSD values increased from initial values of ~1.5 Å to ~2.2 Å at pH 3.5, where they remained stable after ~20 ns, while at pH 5.0 and 6.5 the increase was slower, reaching ~1.4–1.6 Å by the end of the simulations, with transient fluctuations at pH 6.5 between 40 and 70 ns. For HS-tetra(NS), a similar progressive increase was observed at all pH values, with final RMSD values of ~1.6 Å at pH 5.0 and 3.5, while at pH 6.5 the protein remained closer to the reference structure (~1.2 Å). With HS-tetra (NS,2S), backbone RMSD also followed a gradual increase, reaching ~1.4 Å at pH 5.0 and 6.5, but rising more markedly to ~1.8 Å at pH 3.5, with all three systems showing enhanced fluctuations after ~40 ns. These results indicate that the global fold of HPSE is preserved across conditions, but local flexibility increases with simulation time and is modulated by both pH and ligand sulfation.

The RMSD of the ligands relative to their initial docked position showed more pronounced differences between systems (Fig. 7, bottom panels). HS-tetra(0S) remained relatively stable at pH 3.5 (1.4–1.6 Å), but at pH 5.0 it increased rapidly to ~2.2 Å and fluctuated widely between 1 and 3.5 Å, while at pH 6.5 it rose from 0.8 to 1.4 Å after ~65 ns. HS-tetra(NS) displayed initial RMSD values around 2 Å that were maintained at pH 5.0 and 6.5, although with stronger fluctuations at the

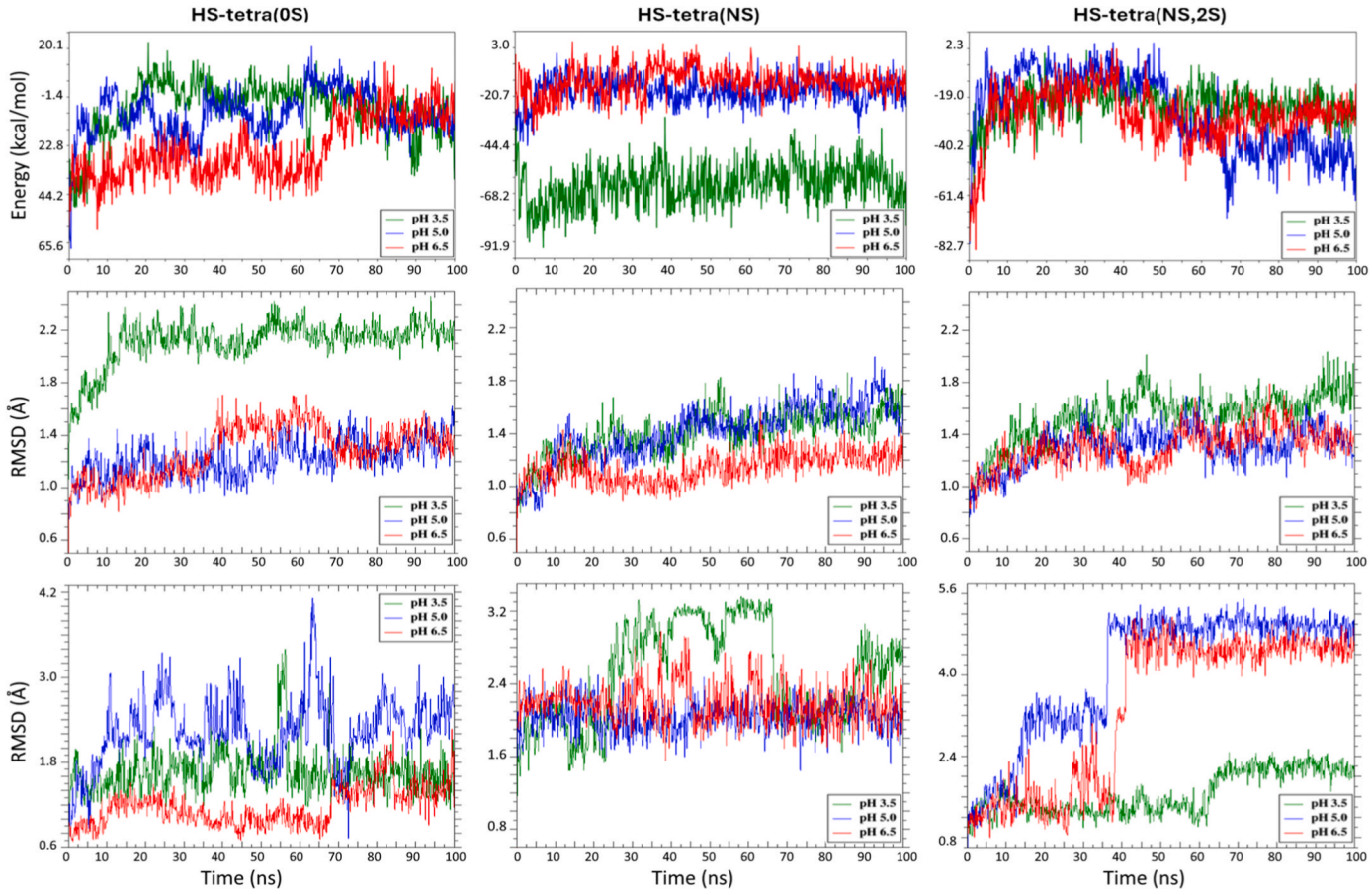


Fig. 6. Molecular dynamics simulation results of HPSE complexes with three heparan sulfate tetrasaccharides [HS-tetra(0S), HS-tetra(NS), and HS-tetra(NS,2S)] at pH 3.5 (green lines), 5.0 (blue lines), and 6.5 (red lines). Top panels: time evolution of the total potential energy of the systems. Middle panels: RMSD of protein Cα atoms during the simulations. Bottom panels: RMSD of the ligands relative to their initial docked position in the binding cleft. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

latter. At pH 3.5, however, a sudden increase to ~3.2 Å occurred after 35 ns, followed by a return to 2 Å around 65 ns, and a subsequent rise to ~2.8 Å at 90 ns, indicating transient rearrangements of the ligand in the binding cleft. The doubly sulfated HS-tetra(NS,2S) exhibited the largest

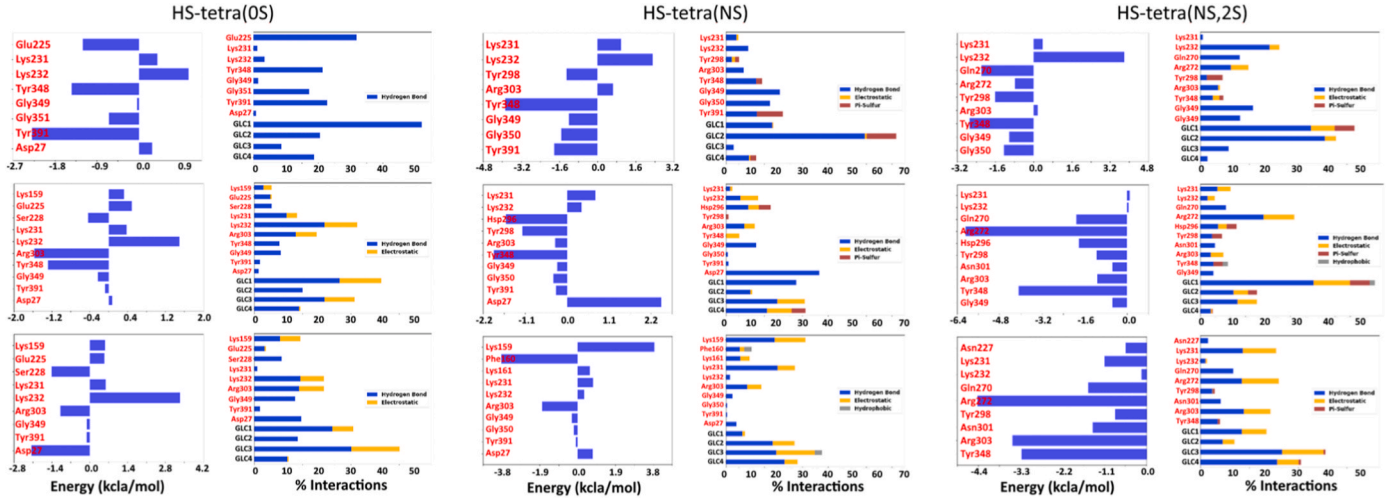


Fig. 7. Residue-level interactions of HPSE with heparan sulfate tetrasaccharides. Each pair of panels shows per-residue energy decomposition (left) and the frequency of interactions (right), including hydrogen bonds (blue), electrostatic interactions (orange), aromatic/π contacts (brown), and hydrophobic contacts (gray). The tetrasaccharides analyzed are HS-tetra(0S) (non-sulfated, left), HS-tetra(NS) (N-sulfated, middle), and HS-tetra(NS,2S) (N- and 2-O-sulfated, right). Rows correspond to pH conditions: top, pH 3.5; middle, pH 5.0; bottom, pH 6.5. Monosaccharides within each tetrasaccharide are labeled GLC1–4 (see Materials and Methods), and the interactions shown represent those primarily mediated by the most engaged monosaccharide(s) in each ligand. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

displacements: at pH 3.5 RMSD increased from 1.2 to ~2 Å after 60 ns, whereas at pH 5.0 and 6.5 abrupt transitions were observed, with RMSD rising from 1.2 Å to ~3.2 Å at 15 ns and further to ~5 Å (pH 5.0) or ~4.5 Å (pH 6.5) after 35–40 ns, where they remained stable. These data suggest that while the protein backbone is relatively insensitive to ligand sulfation, the ligands themselves undergo substantial repositioning depending on their modification pattern and pH. High degrees of sulfation combined with near-neutral pH promote major ligand rearrangements, consistent with enhanced electrostatic repulsion and reduced confinement in the binding pocket, whereas at acidic pH the interactions are more constrained and the ligands remain more stably docked.

Fig. 7 illustrates the residue-level energy decomposition of HPSE in complex with the three tetrasaccharides, together with the frequency of hydrogen bonds (blue), electrostatic interactions (orange), aromatic/π interactions (brown), and hydrophobic contacts (gray). Each row corresponds to a different pH condition: top, pH 3.5; middle, pH 5.0; bottom, pH 6.5. The figure highlights the contribution of individual monosaccharides (GLC1–4) to total interactions, showing how different sulfation patterns and pH conditions influence binding. The monosaccharides GLC1–4 are defined as described in the Materials and Methods. The three heparan-sulfate tetrasaccharides analyzed are:

- HS-tetra(0S): non-sulfated *[GlcNAc-α1,4-GlcUA-β1,4-GlcNAc-α1,4-GlcUA]*
- HS-tetra(NS): N-sulfated at both glucosamine units *[GlcNS-α1,4-GlcUA-β1,4-GlcNS-α1,4-GlcUA]*
- HS-tetra(NS,2S): N-sulfated at glucosamines and 2-O-sulfated at uronic acids *[GlcNS-α1,4-GlcUA-2S-β1,4-GlcNS-α1,4-GlcUA-2S]*

For each tetrasaccharide, the monosaccharides are labeled as follows: GLC1 (first unit), GLC2 (second), GLC3 (third), and GLC4 (fourth).

For the non-sulfated tetrasaccharide HS-tetra(0S), Tyr391 (−2.3 kcal/mol), Tyr298 (−1.5 kcal/mol), Tyr348 (−1.4 kcal/mol), and Glu225 (−1.3 kcal/mol) are the main stabilizing residues, whereas Lys232 frequently contacts the ligand but contributes unfavorably (+1.05 kcal/mol). The first monosaccharide, GLC1 (GlcNAc), mediates the majority of hydrogen bonds (up to 44%). At higher pH, additional residues such as Arg272 (−1.7 kcal/mol), Arg303 (−1.6 kcal/mol), Ser228 (−1.4 kcal/mol), and Asp27 (−2.2 kcal/mol) contribute, and distal monosaccharides (GLC3) become more involved (24% hydrogen-bond + 12% electrostatic interactions), indicating partial reorientation of the ligand.

The N-sulfated tetrasaccharide HS-tetra(NS) exhibits strong interactions with Tyr348 (−4.0 kcal/mol), Tyr391 (−1.8 kcal/mol), Tyr298 (−1.5 kcal/mol), and Thr62 (−1.6 kcal/mol) at acidic pH, with GLC2 (GlcNS) dominating hydrogen-bond interactions (41%) and minor contributions from π/electrostatic contacts. Across pH conditions, residues including Tyr348 (−2.0 kcal/mol), Ala388 (−1.8 kcal/mol), Asn29 (−1.5 kcal/mol), Arg303 (−1.8 kcal/mol), and Phe160 (−3.8 kcal/mol) provide stabilization. At near-neutral pH, distal monosaccharides GLC3 and GLC4 contribute 17–27% of total hydrogen bonds and electrostatic interactions, reflecting a shift of contacts toward the tetrasaccharide termini.

For the doubly sulfated tetrasaccharide HS-tetra(NS,2S), Asp27 (−4.4 kcal/mol), Tyr348 (−2.5 kcal/mol), Tyr391 (−2.4 kcal/mol), and Gln270 (−2.2 kcal/mol) dominate at acidic pH, with GLC1–3 (GlcNS, GlcUA-2S, GlcNS) collectively mediating ~26–27% of hydrogen-bond or π interactions each. At pH 5.0, Arg272 (−6.0 kcal/mol) is the main stabilizer, supported by Tyr348 (−4.0 kcal/mol) and Gln270 (−2.0 kcal/mol), with GLC1 responsible for 36% hydrogen-bond + 10% electrostatic + 5% π + 2% hydrophobic interactions. At pH 6.5, Arg272 (−4.5 kcal/mol), Arg303 (−3.5 kcal/mol), and Tyr348 (−3.2 kcal/mol) maintain strong interactions, with GLC3 mediating 22% hydrogen-bond + 11% electrostatic + 1% π and GLC4 28% hydrogen-bond + 6% electrostatic + 1% π, confirming persistent engagement of the sulfated tetrasaccharide termini.

Overall, ligand sulfation and pH modulate both the distribution and magnitude of HPSE–tetrasaccharide interactions. Acidic conditions favor localized, aromatic- and acidic-residue-mediated binding, while near-neutral pH redistributes contacts toward basic residues and distal monosaccharides.

Fig. 8 illustrates the number of interactions between different heparan sulfate tetrasaccharide ligands and HPSE at three pH values (3.5, 5.0, and 6.5). In the interaction scale, groups contributing to a higher number of interactions are shown in red, while those involved in fewer interactions are shown in green, providing a visual indication of interaction density across the ligands.

In the non-sulfated ligand (HS-tetra(0S)), interactions are primarily mediated by hydroxyl and carboxyl groups of GlcNAc and GlcUA residues. Specifically, the C4 hydroxyl of GlcNAc1 shows the most interactions at pH 6.5, while the C2 and C3 hydroxyls and the carboxyl group of GlcUA4 exhibit substantial interactions across pH 3.5–6.5. The O atom of the β(1 → 4) linkage in GlcUA2 also participates at pH 6.5, indicating that backbone oxygens contribute to stabilizing the ligand–enzyme complex.

Sulfation strongly increases the number of interactions, as observed in HS-tetra(NS) and HS-tetra(NS,2S) ligands. Sulfate groups on GlcNS1 and GlcNS3 consistently contribute the most interactions across all pH values, highlighting their central role in electrostatic recognition by HPSE. Similarly, the 2-O-sulfates and carboxylates of GlcUA-2S2 and GlcUA-2S4 form multiple interactions, particularly at pH 5.0 and 6.5. Hydroxyl groups on GlcNS residues (e.g., C6 OH of GlcNS3) contribute more at lower pH (3.5), reflecting the influence of protonation on hydrogen-bonding capacity.

Overall, these results demonstrate that both sulfation pattern and pH critically influence the interaction profile. Non-sulfated residues rely mainly on hydroxyl and carboxyl groups, whose interactions are more pH-dependent.As can be seen, the affinity of the ligand (mediated by the binding energy) and the frequency of these interactions are favored under acidic conditions (pH 3.5) on the ligand with the highest degree of sulfation. Furthermore, sulfated residues provide multiple contact points, ensuring proper binding across a broader pH range. The data highlight the cooperative contribution of charged (SO_3^-, COO^-) and polar (OH) groups in ligand recognition, and illustrate how specific monosaccharide positions and functional groups determine the strength and distribution of interactions with HPSE.

4. Discussion

The integration of spectroscopic and thermodynamic data indicates that the conformational stability of HPSE is strongly dependent on pH. Importantly, the spectroscopic experiments performed in this study were carried out on the apo form of HPSE and therefore primarily describe global structural stability rather than catalytic activity. FTIR analyses show that acidic conditions promote an increase in ordered secondary-structure elements, accompanied by a decrease in random coil content, suggesting a compaction of the protein scaffold. These features are consistent with the reported optimal activity of HPSE near lysosomal pH [56,57], although these measurements refer to the apo enzyme. Fluorescence-based melting measurements reveal maximal stability at pH 5.0 and a progressive decrease upon approaching neutrality [58]. The absence of cooperative transitions indicates a gradual denaturation process consistent with an equilibrium among multiple conformational states. Overall, these results suggest that HPSE adopts a more ordered and thermally stable structure under mildly acidic conditions.

The global stability differences observed experimentally are reflected locally in the active-site microenvironment. MD simulations indicate that at pH values close to the lysosomal range, the protein exhibits reduced local flexibility, particularly in loops surrounding the binding pocket. This reduced mobility may favor structural arrangements compatible with catalytic geometry and could help rationalize the

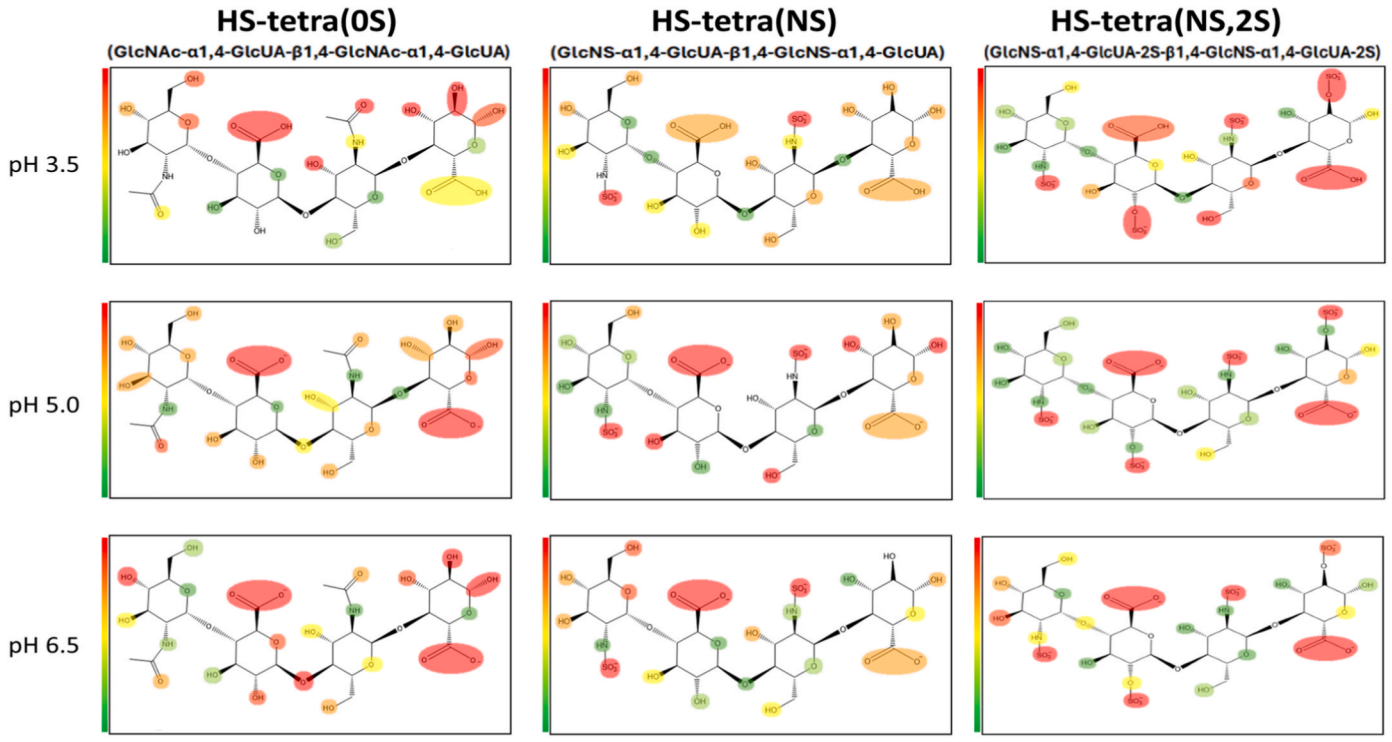


Fig. 8. Interaction maps of heparan sulfate tetrasaccharide ligands with HPSE at pH 3.5, 5.0, and 6.5 (from top to bottom). Red areas indicate groups contributing the highest number of interactions, while green areas correspond to regions with fewer interactions. Columns represent ligands with increasing sulfation (0S, NS, and NS,2S). The color scale, shown to the left of each panel, ranges from green (fewer interactions) to red (more interactions). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

higher enzymatic efficiency reported under acidic conditions [59]; however, this interpretation remains indirect, as no functional measurements were performed in this study. Conversely, at near-neutral pH, enhanced loop dynamics may impair the optimal residue alignment for substrate recognition.

The secondary structure changes observed by FTIR primarily reflect global conformational rearrangements that may influence the dynamic properties of regions involved in substrate recognition. Increases in α-helix and β-sheet content may correspond to reduced flexibility in structural elements contributing to substrate positioning, thereby affecting geometry and accessibility of the binding pocket.

With regard to substrate recognition, the simulation experiments show that sulfated tetrasaccharides are stabilized by a complex balance of ionic interactions, hydrogen bonds, and dynamic contributions. Increasing ligand sulfation is associated with more favorable interaction energies and more stable RMSD profiles over time, indicating greater ligand confinement within the binding cavity. In this context, stability describes the persistence of protein-ligand interactions and reduced positional fluctuations rather than the absence of local rearrangements. This effect is more pronounced under acidic conditions and suggests that highly sulfated ligands form more persistent interaction networks that contribute to ligand retention within the binding site. Moreover, complex formation induces local stabilization of surrounding loops, suggesting an interplay between active-site organization and substrate sulfation degree. This molecular picture is consistent with the experimentally observed selectivity of HPSE for highly sulfated regions of heparan sulfate [35]. However, since the ligands considered here mainly differ in sulfation density while sharing a similar backbone structure, the simulations primarily highlight sulfation- and electrostatics-driven contributions rather than the full sequence specificity of natural substrates.

Overall, these results provide insights into the structural behavior of HPSE under different pH conditions. While the enzyme exhibits optimal activity around pH 5.0 [60,61], the present spectroscopic measurements mainly reflect its structural stability. The data indicate that near-neutral pH conditions promote partial structural destabilization, whereas excessively acidic conditions may reduce structural flexibility. These observations suggest that pH-dependent structural changes may influence the conformational dynamics of the protein, although direct functional measurements would be necessary to confirm their impact on catalytic activity.

These findings are relevant in the physiological and pathological context of HPSE action. The preference for acidic conditions reflects the lysosomal environment where HS degradation occurs. The selectivity for highly sulfated regions is consistent with the HPSE roles in extracellular matrix remodeling, growth factor release, and cell migration [62]. In pathological contexts such as metastasis or chronic inflammation, HPSE activity on densely sulfated HS domains contributes to modulating signaling pathways and invasive processes [63,64]. The obtained results may therefore help to clarify how variations of pH or HS composition influence these biological processes and may provide a structural basis for the future development of inhibitors targeting electrostatic interactions or protein dynamics.

Taken together, these findings provide a coherent structural framework for understanding the physicochemical factors modulating HPSE stability and substrate interactions, which may be further refined by future experimental and computational studies.

5. Conclusions

In this study, we combined spectroscopic and computational approaches to characterize the influence of pH and substrate sulfation on the structural stability and substrate recognition of HPSE. FTIR analyses show that acidic conditions promote a more compact conformation of the enzyme, characterized by an increase in α-helical and β-sheet structures. This secondary-structure reorganization is consistent with the lysosomal environment in which HPSE is physiologically active. Intrinsic fluorescence measurements further indicate that thermal

unfolding proceeds gradually, without sharp cooperative transitions, outlining a continuum of conformational states. Melting temperatures suggest that HPSE reaches maximal structural stability under mildly acidic conditions, with a marked decrease in stability near neutrality.

Molecular dynamics simulation experiments provide a complementary view of the conformational flexibility of the enzyme and the structural determinants of substrate recognition. Highly sulfated heparan-sulfate tetrasaccharides form stronger and more persistent ionic and hydrogen-bond interactions with key active-site residues (including Lys159, Arg272, and Arg303), in agreement with the experimentally observed preference of HPSE for highly sulfated domains. The simulation experiments also suggest that the increasing sulfation density is associated with more favorable interaction energies and reduced ligand mobility within the binding pocket.

Overall, the integration of spectroscopic measurements and computational simulation outlines a coherent picture in which mildly acidic pH favors a more ordered and thermally stable structure of HPSE, while higher ligand sulfation promotes more persistent enzyme-substrate interactions. These findings contribute to a better understanding of the physicochemical factors that may influence HPSE stability and substrate binding under physiological and pathological conditions and may support future efforts aimed at the development of inhibitors or modulators targeting this enzyme.

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CRediT authorship contribution statement

Angela Pennacchio: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization, Writing – review & editing. **Yesid Aristizabal:** Data curation, Formal analysis, Investigation, Software, Visualization. **José Oñate-Garzón:** Funding acquisition, Resources, Validation, Writing – review & editing. **Sabato D'Auria:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Alessio Ausili:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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During the preparation of this work the author(s) used ChatGPT to improve language, grammar, and clarity of the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

No data was used for the research described in the article.

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